Agentic Software Engineering (SE 3.0): The Rise of AI Teammates

Hao Li*
Queen's University
Kingston, ON, Canada
hao.li@queensu.ca

Haoxiang Zhang
Queen's University
Kingston, ON, Canada
haoxiang.zhang@queensu.ca

Jie M. Zhang
King's College London
London, UK
jie.zhang@kcl.ac.uk

Yiling Lou
University of Illinois
Urbana-Champaign
Urbana, IL, USA
yilingl@illinois.edu

Baishakhi Ray
Columbia University
New York, NY, USA
rayb@cs.columbia.edu

Thomas Zimmermann
University of California, Irvine
Irvine, CA, USA
tzimmer@uci.edu

Ahmed E. Hassan
Queen's University
Kingston, ON, Canada
ahmed@cs.queensu.ca

Abstract

We are witnessing the emergence of Agentic Software Engineering (SE 3.0), where AI agents act as autonomous AI Teammates performing complex tasks such as coding, debugging, and testing. As AI Teammates generate a vast new category of digital artifacts, they introduce unique opportunities and challenges related to human-AI collaboration, trustworthiness, and economic impact. This workshop serves as the premier forum for addressing these challenges, anchored by the launch of the AIDev dataset. Comprising over one million agentic pull requests generated by AI Teammates such as Claude Code, OpenAI Codex, and GitHub Copilot, AIDev provides the empirical evidence needed to understand the behaviors of AI Teammates. This workshop features insights from major industry players and academic pioneers, and aims to define a roadmap for a world where AI Teammates and human developers build the future together.

ACM Reference Format:
Hao Li, Haoxiang Zhang, Jie M. Zhang, Yiling Lou, Baishakhi Ray, Thomas Zimmermann, and Ahmed E. Hassan. 2026. Agentic Software Engineering (SE 3.0): The Rise of AI Teammates. In *Proceedings of Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '26).* ACM, New York, NY, USA, 2 pages. https://doi.org/XXXXXXX.XXXXXXX

*Main contact. Phone: +1-343-364-5663

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
KDD '26, Jeju, Korea
© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-x-xxxx-xxxx-x/YYYY/MM
https://doi.org/XXXXXXX.XXXXXXX

1 Workshop Overview

Topics. Software engineering is evolving from a human-centric discipline into a collaboration between humans and AI agents, i.e., AI Teammates [1]. The Agentic Software Engineering (SE 3.0) workshop targets the emerging field of agentic software development and the massive data trails generated by these emerging AI Teammates. As AI Teammates move beyond simple code completion to performing complex tasks such as fixing bugs, testing code, and managing pull requests (PRs), they generate a vast new category of digital artifacts. To provide an empirical foundation for this shift, we introduce the AIDev dataset [2], a collection of over one million agentic Pull Requests authored by tools like Claude Code, GitHub Copilot, and OpenAI Codex. We aim to explore the data mining and knowledge discovery challenges inherent in these workflows, with a focus on understanding the dynamic behaviors, reasoning patterns, limitations, and collaborative impacts of AI Teammates within the global software ecosystem. The workshop covers the following key topics:

- Mining agent-generated artifacts
- Human-agent collaboration and alignment
- Trustworthiness, assurance, and reliability
- Multi-agent coordination and communication
- Agent tool use and environments
- Failure modes and root causes
- Safety, ethics, and compliance
- Adoption and productivity
- Economic cost and impact
- AgentOps and release engineering

2 Motivation and Rationale

We are currently at a historic tipping point in software development, marked by the transition from a human-centric coding model to one in which agentic coders (e.g., Claude Code, Cursor, Devin, GitHub Copilot, and OpenAI Codex) are becoming a serious, high-volume workforce. With over one million PRs on GitHub already

attributed to AI Teammates, the AIDev [2] large-scale dataset is no longer theoretical but an industrial reality. We must act now to understand the cost and impact of these workflows to ensure human-agent alignment, reduce the cognitive workload assigned to human reviewers, and move toward a future in which human intent and agentic execution jointly deliver high-value software with maximum efficiency.

3 Organization

3.1 Program Sketch

The workshop will be a half-day event (~3 hours).

3.2 Timeline

We will follow the suggested KDD 2026 Workshop timeline:

- Workshop Website and CFP: April 9th, 2026
- Workshop Paper Submission: April 30th, 2026
- Workshop Paper Notification: June 4th, 2026
- Final Program Online: June 22nd, 2026

4 Organizers

Hao Li is a Postdoctoral Fellow at Queen's University. He served as the Co-Chair of the MSR 2026 Mining Challenge. His research investigates the intersection of Software Engineering and AI, with a focus on agentic workflows. He is an active member of the research community, serving on the Program Committees for ICSE, FSE, ASE, ICSME, SANER, and AIware.

Haoxiang Zhang is an Adjunct Research Associate at Queen's University. He specializes in mining large-scale software repositories and building AI-driven analytics to improve software quality and reliability. He served as the Co-Chair of the MSR 2026 Mining Challenge.

Jie M. Zhang is a lecturer of computer science at King's College London and a visiting AI scientist at Mistral AI. Her main research interests are AI and software trustworthiness. She is a steering committee member of IEEE ICST and ACM AIware. She is also PC co-chair for ASE 2027 and ISSRE 2026, and area chairs for ASE 2026&2025 and ICSE 2026. Over the last three years, she has been invited to give over 40 talks at conferences, universities, and IT companies. In recognition of her influence, she was named one of the Top 15 Global Chinese Female Young Scholars in Interdisciplinary AI (2023). Her research has won the FSE 2025 distinguished paper award, the 2024 and 2022 Transactions on Software Engineering Best Paper awards, AAAI 2026 oral paper, and ICLR 2022 spotlight paper. She is also the winner of 2025 ACM Sigsoft Early Research Award, one of the most prestigious honours for early-career researchers in the software engineering community.

Yiling Lou is an Assistant Professor in the Siebel School of Computing and Data Science at the University of Illinois Urbana-Champaign. Her research lies at the intersection of software engineering and artificial intelligence, with a focus on LLMs for code, software agents, and trustworthiness of agents. Her work has received multiple ACM SIGSOFT and IEEE TCSE Distinguished Paper Awards. She has served as Program Co-Chair for LLM4Code (2024–2026) and AIware (2025).

Baishakhi Ray is an Associate Professor of Computer Science at Columbia University. She works at the intersection of AI, Software Engineering, and Security. She received her Ph.D. degree from the University of Texas, Austin. She has received the IEEE TCSE Rising Star, NSF CAREER, IBM faculty, and VMWare Faculty awards. She is also an Amazon Visiting Academic. Her research has been acknowledged by many Distinguished Paper awards including FSE'17, ASE'22, ISSTA'23, the ICSME Most Influential Paper award, and has also been published in CACM Research Highlights and has been widely covered in trade media.

Thomas Zimmermann is a Chancellor's Professor and Bren Chair at the University of California, Irvine, where he leads research at the intersection of software engineering, machine learning, artificial intelligence, and data science. With more than 20 years of experience and over 100 publications cited more than 30,000 times, his work has shaped how people understand and improve software development. His research has informed and improved major Microsoft products and tools (Visual Studio, GitHub, and Xbox) during his tenure at Microsoft Research (2008-2024). He is a Fellow of the ACM, IEEE and AAAS Fellow as well as a recipient of the IEEE TCSE Edward J. McCluskey Technical Achievement Award. More information is available at http://thomas-zimmermann.com.

Ahmed E. Hassan is a Mustafa Prize Laureate and Fellow of ACM, IEEE, and AAIA, and an NSERC Steacie Fellow, he holds the Canada Research Chair and the NSERC/BlackBerry Industrial Research Chair in Software Engineering (SE) at Queen's University and is among the world's most cited SE researchers. A founder of the AI-Augmented SE, MSR, and AIware communities and a member of the Royal Society of Canada, he has led in industry (IBM Almaden, BlackBerry) and academia for three decades. He is the only person to receive both the ACM SIGSOFT Influential Educator Award (2019) and the IEEE TCSE Distinguished Educator Award (2020).

References

[1] Ahmed E. Hassan, Hao Li, Dayi Lin, Bram Adams, Tse-Hsun Chen, Yutaro Kashiwa, and Dong Qiu. 2025. Agentic Software Engineering: Foundational Pillars and a Research Roadmap. arXiv:2509.06216 [cs.SE] https://arxiv.org/abs/2509.06216

[2] Hao Li, Haoxiang Zhang, and Ahmed E. Hassan. 2025. The Rise of AI Teammates in Software Engineering (SE) 3.0: How Autonomous Coding Agents Are Reshaping Software Engineering. arXiv:2507.15003 [cs.SE] https://arxiv.org/abs/2507.15003